

B.E. – Electrical and Electronics Engineering

Total Credits to be earned for the award of the Degree: 165 (Regular); 125 (Lateral Entry)

R2023 BE-EEE Curriculum

SEMESTER I											
Course Code		Course Name	L	T	P	H	C	Maximum Marks			Category
								CA	ES	Total	
Theory Course											
	2301IP101	Induction Program	0	0	0	0	0	0	0	0	
1	2301MA105	Engineering Mathematics – I (Matrices and Ordinary Differential Equations)	3	2	0	5	4	40	60	100	BSC
2	2301TA101	Heritage of Tamils/தமிழர் மரபு	1	0	0	1	1	100	0	100	HSMC
3	2301PH103	Semiconductor Physics and Optoelectronics	3	0	0	3	3	40	60	100	BSC
4	2301GEX02	Engineering Graphics	2	2	0	4	3	40	60	100	ESC
5	2301ENX01	Professional English	2	0	0	2	2	40	60	100	HSMC
Practical Course											
6	2301PHX51	Engineering Physics Laboratory	0	0	2	2	1	60	40	100	BSC
7	2301GEX52	Engineering Practices Laboratory	0	0	4	4	2	60	40	100	ESC
8	2301ENX51	Communication Skills Laboratory	0	0	2	2	1	100	0	100	HSMC
9	2301LS101	Life Skill Activity - I	0	0	0	0	0	100	0	100	-
Total			11	02	08	23	17	580	320	900	

SEMESTER II											
Course Code	Course Name	L	T	P	H	C	Maximum Marks			Category	
							CA	ES	Total		
Theory Course											
1	2301MA205	Engineering Mathematics –II (Calculus and Computational Methods)	3	2	0	5	4	40	60	100	BSC
2	2302EE201	Electric Circuit Analysis	3	0	0	3	3	40	60	100	PCC
3	2301GE204	Basic Civil and Mechanical Engineering	3	0	0	3	3	40	60	100	ESC
4	2301TA201	Tamils and Technology/ தமிழரும் தொழில்நுட்பமும்	1	0	0	1	1	100	0	100	HSMC
5	2301CH202	Applied Chemistry	3	0	0	3	3	40	60	100	BSC
6	--	Foreign Language	2	0	0	2	2	100	0	100	EEC
Theory Cum Laboratory Course											
7	2301GEX04	Problem Solving using Python	2	0	4	6	4	50	50	100	ESC
Practical Course											
8	2301CHX51	Engineering Chemistry Laboratory	0	0	2	2	1	60	40	100	BSC
9	2301GE252	Mathematics Simulation Laboratory	0	0	2	2	1	60	40	100	BSC
10	2301GEX51	Computer Practices Laboratory	0	0	2	2	1	60	40	100	ESC
11	2301LS201	Life Skill Activity - II	0	0	0	0	0	100	0	100	-
Total			17	02	10	29	23	690	410	1100	

SEMESTER III											
Course Code		Course Name	L	T	P	H	C	Maximum Marks			Category
								CA	ES	Total	
Theory Course											
1	2301MA305	Engineering Mathematics-III (Transforms and Optimization)	3	2	0	5	4	40	60	100	BSC
2	2302EE301	Electrical Machines-I	3	2	0	5	4	40	60	100	PCC
3	2302EE302	Analog Electronics	3	0	0	3	3	40	60	100	PCC
4	2302EE303	Digital Electronics	3	0	0	3	3	40	60	100	PCC
Theory Cum Laboratory Course											
5	2302EE304	Measurements & Instrumentation	3	0	2	5	4	50	50	100	PCC
6	2301GEX07	Environmental Science and Sustainability	1	0	2	3	2	50	50	100	BSC
Practical Course											
7	2302EE351	Electrical Machines–I Laboratory	0	0	2	2	1	60	40	100	PCC
8	2302EE352	Analog and Digital Electronics Laboratory	0	0	2	2	1	60	40	100	PCC
9	2304GE301	Professional Development -1	0	0	2	2	1	100	0	100	EEC
10	2301LS301	Life Skill:03	0	0	0	0	0	100	0	100	-
Total			16	04	10	30	23	580	420	1000	

SEMESTER IV											
Course Code	Course Name	L	T	P	H	C	Maximum Marks			Category	
							CA	ES	Total		
Theory Course											
1	2302EE401	Generation, Transmission and Distribution	3	2	0	5	4	40	60	100	PCC
2	2302EE402	Electrical Machines -II	3	2	0	5	4	40	60	100	PCC
3	2302EE403	Power Electronics	3	0	0	3	3	40	60	100	PCC
4	2302EE404	Linear Control Systems	3	2	0	5	4	40	60	100	PCC
5	2301MC401	Mandatory course-I (Design Thinking For Innovation)	2	0	0	2	0	100	0	100	MC
Theory Cum Laboratory Course											
6	2302EE405	Linear Integrated Circuits	3	0	2	5	4	50	50	100	PCC
7	2301HSX01	Universal Human Values and Ethics	1	0	2	3	2	50	50	100	HSSMC
Practical Course											
8	2302EE451	Electrical Machines –II Laboratory	0	0	2	2	1	60	40	100	PCC
9	2302EE452	Power Electronics and Drives Laboratory	0	0	2	2	1	60	40	100	PCC
10	2304GE401	Professional Development -2	0	0	2	2	1	100	0	100	EEC
11	2301LS401	Life Skill :04	0	0	0	0	0	100	0	100	-
Total			18	06	10	34	24	680	420	1100	

SEMESTER V											
Course Code		Course Name	L	T	P	H	C	Maximum Marks			Category
								CA	ES	Total	
Theory Course											
1	2302EE501	Microprocessor and Microcontroller	3	0	0	3	3	40	60	100	PCC
2	2302EE502	Power System Analysis	3	2	0	5	4	40	60	100	PCC
3	-	Elective –I	3	0	0	3	3	40	60	100	PEC
4	-	Elective –II	3	0	0	3	3	40	60	100	PEC
5	-	Open Elective -I	3	0	0	3	3	40	60	100	OEC
Theory Cum Laboratory Course											
6	2302CS505	Java Programming	2	0	2	4	3	50	50	100	PCC
Practical Course											
7	2302EE551	Computer Aided Electrical Drawing Laboratory	0	0	2	2	1	60	40	100	PCC
8	2302EE552	Microprocessor and Microcontroller Laboratory	0	0	2	2	1	60	40	100	PCC
9	2304GE501	Professional Development -3	0	0	2	2	1	100	0	100	EEC
10	2301LS501	Life Skill: 05	0	0	0	0	0	100	0	100	-
Total			17	02	08	27	22	570	430	1000	

SEMESTER VI											
Course Code		Course Name	L	T	P	H	C	Maximum Marks			Category
								CA	ES	Total	
Theory Course											
1	2302EE601	Solid State Drives	3	0	0	3	3	40	60	100	PCC
2	2302EE602	Protection and Switchgear	3	2	0	5	4	40	60	100	PCC
3	-	Elective – III	3	0	0	3	3	40	60	100	PEC
4	-	Elective –IV	3	0	0	3	3	40	60	100	PEC
5	-	Open Elective -II	3	0	0	3	3	40	60	100	OEC
6	-	Mandatory course-II	2	0	0	2	0	100	0	100	MC
Theory Cum Laboratory Course											
7	2302EE603	Embedded Systems	3	0	2	5	4	50	50	100	PCC
Practical Course											
8	2302EE651	Power Systems Laboratory	0	0	2	2	1	60	40	100	PCC
9	2304EE652	Mini-Project	0	0	2	2	1	50	50	100	EEC
10	2304GE601	Professional Development – 4	0	0	2	2	1	100	0	100	EEC
11	2301LS601	Life Skill : 06	0	0	0	0	0	100	0	100	-
Total			20	02	08	30	23	660	440	1100	

SEMESTER VII											
Course Code		Course Name	L	T	P	H	C	Maximum Marks			Category
								CA	ES	Total	
Theory Course											
1	2302EE701	Utilization of Electrical Energy	3	0	0	3	3	40	60	100	PCC
2	2302EE702	Power System Operation and control	3	0	0	3	3	40	60	100	PCC
3	2302EE703	High Voltage Engineering	3	0	0	3	3	40	60	100	PCC
4	-	Elective - V	3	0	0	3	3	40	60	100	PEC
5	-	Elective- VI	3	0	0	3	3	40	60	100	PEC
6	-	Open Elective -III	3	0	0	3	3	40	60	100	OEC
Theory Cum Laboratory Course											
7	2302EE704	Renewable Energy Systems	3	0	2	5	4	50	50	100	PCC
Practical Course											
8	2304EE751	Project phase – I	0	0	4	4	2	100	0	100	EEC
Total			21	0	6	27	24	390	410	800	

SEMESTER VIII										
Course Code		Course Name	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
Practical Course										
1	2304EE851	Project phase – II	0	0	16	08	50	50	100	EEC
2	2304EE852	Internship / Industrial Visit/ In plant Training Presentation	0	0	2	01	100	00	100	EEC
	Total		0	0	18	09	150	50	200	

Mandatory course offered by Department	Mandatory course-I	Mandatory course-II
	2301MC401 - Design Thinking for Innovation	2301MC613 - Mental Health & Wellbeing

Open Elective course offered by Department	Open Elective-I (Odd Sem)	Open Elective-II (Even Sem)
	2303EE029 - Energy Conservation and Energy Management	2303EE031 - Electronic Waste Management Issues and Challenges

PROFESSIONAL ELECTIVE COURSES: VERTICALS

Electives	I - Electric Vehicles	II - Smart Energy Grid	III - Embedded Systems	IV - Diversified Courses
Elective-I	2303EE001 - Fundamentals of Electric vehicle and its architecture	2303EE008 - Restructured Power System and its Market	2303EE015 - Embedded Programming	2303EE022 - Special Electrical Machines
Elective-II	2303EE002 - Energy Storage and Battery Management System	2303EE009 - Power Quality	2303EE016 - Micro Controller Based System Design	2303EE023 - Electrical Machine Design
Elective-III	2303EE003 - Design of motor and power converters for electric vehicles	2303EE010 - Distributed generation system	2303EE017 -VLSI Design	2303EE024 - Electricity markets
Elective-IV	2303EE004 - Design of Electric Vehicle Charging Systems	2303EE011 - SCADA and Substation Automation	2303EE018 - Smart system Automation	2303EE025 - Flexible AC Transmission Systems
Elective-V	2303EE005 - Testing of Electric Vehicles and Standards	2303EE012 - Smart Grid Technologies	2303EE019 - ML and IoT Frameworks	2303EE026 - Optimization Techniques
Elective-VI	2303EE006 - Grid Integration of Electric Vehicles	2303EE013 - AI and ML Techniques for Power Systems	2303EE020 - Industry 4.0 and Industrial Internet of things	2303EE027 - HVDC Transmission
	2303EE007 - Autonomous Electric Vehicles	2303EE014 - Data Analytics and Data Driven Decision making for modern power system	2303EE021 - Advance Topics in Embedded AI	2303EE028 - MEMS and NEMS

OPEN ELECTIVE COURSES (OEC)

Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CA	ES	Total	
2303EE029	Energy Conservation and Energy Management	3	0	0	3	40	60	100	OEC
2303EE030	Troubleshooting of Electrical Appliances	2	0	2	3	50	50	100	OEC
2303EE031	Electronic Waste Management Issues and Challenges	3	0	0	3	40	60	100	OEC
2303EE032	Electrical Drives and Control	3	0	0	3	40	60	100	OEC
2303EE033	Foundations of Electric Vehicles	3	0	0	3	40	60	100	OEC